

Digital Signal Analysis and Processing (CT704)

Detailed PYQ

July 22, 2026

Notes

- Questions from Tribhuvan University, IOE (2069–2081 BS), BEI/BCT, single code CT704.
- Regular exam = **bold**, Back exam = regular font.
- Months: Bh = Bhadra, Ba = Baishakh, Ash = Ashwin/Ashar, Shr = Shrawan, Ch = Chaitra, Ka = Kartik.
- Keywords: (I) = Importance/Need, (+) = Advantage, (-) = Disadvantage, +Fig = Figure, +Eg = Example, Vs = Compare/Differentiate.
- Marks kept in [brackets] as originally split (e.g. [2+6]).

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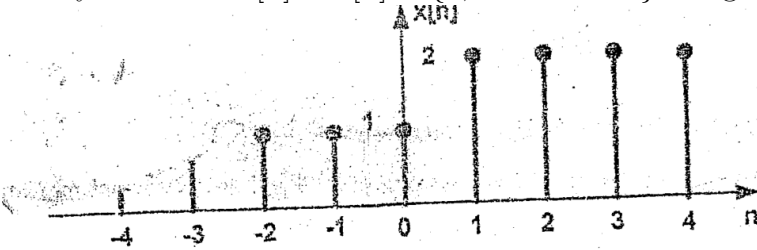
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1 Discrete-Time Signals and Systems

(8 Hours)

1.1 Basic signals, even/odd, transformation of variable

1. Define even and odd type discrete time signals with example. Plot the signal $x[-2n + 3]$ where $x[n] = \{1, 2, 0, -1, -3, -4\}$. [2+3] (76 Ch)
2. Find the even and odd part of the signal $x[n] = \{1 \text{ for } -4 \leq n \leq 0; 2 \text{ for } 1 \leq n \leq 4\}$. [3] (71 Ch, 70 Ash)
|→ Find the odd and even part of the signal shown in the stem plot below, and find the output of an LTI system with $x[n] = h[n] = \{1, -1 \leq n \leq 1\}$ using the graphical method. [4+5] (71 Shr)



3. Define and plot a discrete time unit step signal. Explain its relation with the unit impulse signal. [1+2] (73 Ch)

1.2 Energy / power signals and periodicity

1. Define (compare) energy and power signal, and determine whether the given signal is periodic (find its fundamental period).
|→ $x[n] = \cos(2\pi n/5) + \sin(\pi n/3)$ [2+2] (79 Bh, 69 Ch)
|→ $x[n] = e^{j(\pi n/3 + \pi/4)}$ [2+2] (74 Ash)
|→ $x[n] = e^{j(\pi n/2 + 4\pi/7)}$ (energy or power?) [2+2] (79 Ba)
|→ $x[n] = u[n]$ and $x[n] = \delta[n]$: energy or power type? [2+3] (72 Ka)
|→ Power Vs energy signal; Fourier Series Vs Fourier Transform. [3+4] (75 Ch)
2. Determine whether the signal is periodic; if periodic, state its periodic/fundamental time.
|→ $x[n] = e^{j\pi n/16} \cos(n\pi/17)$ [5] (80 Bh)
|→ (a) $\sin(n\pi) + \cos(n\pi)$ (b) $\sin(3n\pi/5) + \cos(4n\pi/7)$ [2+2] (80 Ba)
|→ $x[n] = \cos(\pi n/2) \cos(\pi n/4)$ [4] (78 Bh)
|→ (i) $\cos(\pi n^2/8)$ (ii) $\cos(n/2) \cos(\pi n/4)$ [3] (70 Ch)

1.3 System properties (linearity, time-invariance, causality, stability)

1. A DT system is given by $y(n) = \sum_{k=0}^n x(k)$; check whether it is memoryless, time-invariant and stable. [2+2+2] (80 Bh)
2. Determine the given systems: (a) $y[n] = x[-n]$ time-invariant? (b) $y[n] = x[n^2]$ linear? [5] (76 Ch)
|→ (a) $y[n] = y[n-4] + x[n-4]$ time-invariant? (b) $y[n] = x^2[n]$ linear? [3+3] (74 Ch)
|→ (a) $y[n] = x^2[n]$ (b) $y[n] = \cos(\frac{5\pi}{8}n + \frac{\pi}{4})$: linear or not? [3+3] (75 Ash)
3. An LTI system is given by $y(n) = x(n) + e^a y(n-1)$; check this system for BIBO stability. [5] (81 Ba)

1.4 Fourier series/transform of DT signals; convolution

1. Explain the process of calculating Fourier series coefficients. [3] (73 Shr) [4] (72 Ch)

2. Explain the Fourier transform multiplication property for two sequences. Write Dirichlet conditions for Fourier series. [4+3] (76 Ash)
3. Find the output $y(n)$ of an LTI system given its impulse response h and input x (convolution). Representative parameter sets:
 - $\rightarrow h(n) = \{5, 4, 3, 2\}, x(n) = \{1, 0, 3, 2\}$ [6] (70 Ch)
 - $\rightarrow h(n) = \{1, 1, 1\}, x(n) = \{1, 1, 1, 1\}$ [6] (73 Shr)
 - $\rightarrow h[n] = (1/2)^n \{u[n+2] - u[n-2]\}, x[n] = \{2, 1, 0, -1, 4\}$ [5] (79 Ba)
 - $\rightarrow h[n] = (1/2)^n \{u[n+2] - u[n-2]\}, x[n] = \{2, 1, 0.5, -1\}$ [3+2] (74 Ash)
 - $\rightarrow h[n] = (1/3)^n \{u[n+1] - u[n-2]\}, x[n] = \{2, 1, 0.5, 3\}$ [5] (72 Ka)
 - $\rightarrow h[n] = u[n] - u[n-4], x[n] = (1/2)^n u[n]$ [5] (78 Bh)
 - $\rightarrow h[n] = (1/3)^n u[n], x[n] = 5e^{j\pi n/2}$ [4] (70 Ch)
 - $\rightarrow h[n] = (1/2)^n u[n], x[n] = 5e^{j\pi n/3}$ [4/5] (72 Ka, 80 Ba)
 - $\rightarrow h[n] = (1/2)^n u[n], x[n] = 10 - 5\sin(\pi n/2) + 20\cos(\pi n)$ [3+7] (73 Shr)
 - $\rightarrow h[n] = (1/3)^n u[n-1]/\dots, x[n] = u[n+1] - u[n-4]$ [6] (76 Ch)
 - $\rightarrow x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3], h[n] = 2\delta[n+1] + 2\delta[n-1]$ [5] (79 Bh)
 - $\rightarrow h(n) = \{1, 3, 2, -1, 1\}, x(n) = 2\delta(n) - \delta(n-1)$ [6] (71 Ch)
 - $\rightarrow h(n) = (1/3)^n u[n], x(n) = \{2, 1, 0.5, 3\}$ etc.
4. LTI output from table values (convolution) — $x[-2] = 0.5, x[0] = 1, x[1] = 0.75, x[3] = 0.5; n[0] = 1, n[1] = 0.75, n[2] = 0.5$; verify. [6] (73 Ch)
 - $\rightarrow h[-2] = 3, h[0] = 2, h[1] = 1, x[n] = 2^n, -1 \leq n \leq 3$; check. [5+2] (75 Ch)
 - $\rightarrow h[-2] = 1, h[0] = 2, h[1] = 3, x[0] = 1/2, x[2] = 2, x[3] = 3$; check. [3+2] (72 Ch)
5. Illustrate the significance of convolution summation in digital signal analysis and compute the convolution $h(n) = \{1, 0, 1\}, x(n) = \{1, -2, -2, 3, 4\}$. [2+4] (70 Ash)
 - $\rightarrow$ Find the convolution $x[n] = a^n u[-n], 0 < a < 1, h[n] = u[n]$. [6] (76 Ash)

2 Z-Transform

(4 Hours)

2.1 ROC — definition and properties

1. Define ROC / state and explain the properties of Region of Convergence (and locate the ROC of a signal). [1-3] (**80 Bh,73 Ch,72 Ch**, 79 Ba, 72 Ka, 71 Shr)
 $\rightarrow$ Properties of ROC + locate ROC of $x[n] = (-1/3)^n u[n] - (1/3)^n u[-n-1]$ [2+4] (**70 Ch**)
 $\rightarrow$ Properties of ROC + locate ROC of $x[n] = (0.6)^n u[n] + (0.25)^n u[n]$ [3+6] (75 Ash)
 $\rightarrow$ Properties of ROC + locate ROC of $x[n] = (0.1)^n u[n] + (0.3)^n u[-n-1]$ [4+6] (**75 Ch**)
 $\rightarrow$ List properties of ROC + Z-transform & locate ROC of $x[n] = (-1/3)^n u[n] - (1/3)^n u[-n-1]$ [2+4] (70 Ash)

2.2 Properties of Z-transform (convolution, time-shift)

1. State the convolution property of Z-transform (and derive it). [3+3/3+6] (76 Ash, 72 Ka, 70 Ash)
2. Derive the time shifting property of Z-transform. [3+3] (**69 Ch**)

2.3 Inverse Z-transform (partial fraction / long division)

1. Determine the inverse Z-transform (state the ROC region as given):
 $\rightarrow H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - 0.75z^{-1} + 0.125z^{-2}}, 0.25 < |z| < 0.5$ [1+6] (**80 Bh**)
 $\rightarrow X(z) = \frac{1}{1 - 0.8z^{-1} + 0.12z^{-2}}$ for $|z| > 0.6, |z| < 0.2, 0.2 < |z| < 0.6$ [2+2+3] (81 Ba)
 $\rightarrow X(z) = \frac{1}{1 + 2z^{-1} + z^{-2}}, |z| > 1$ [1+5] (80 Ba)
 $\rightarrow X(z) = \frac{2z^4 + 2z^3 - 3z + 2}{z^2 - 1.5z - 1}, |z| < 0.5$ (partial fraction) [6] (**79 Bh**)
 $\rightarrow H(z) = \frac{z}{3z^2 - 4z + 1}, \frac{1}{3} < |z| < 1$ (partial fraction) [1+5] (79 Ba)
 $\rightarrow X(z) = \frac{z^3 + z^2 + 1.5z + 0.5}{z^3 + 1.5z^2 + 0.5z}, |z| < 1/2$ [1+5] (**78 Bh**)
 $\rightarrow X(z) = \frac{z^4 + 5z^3 - 3z + 4}{z^2 - 1.5z - 1}, |z| < 0.5$ [2+6] (**76 Ch**)
 $\rightarrow X(z) = \frac{z}{(z - 0.6)(z + 0.5)^2}, |z| > 0.6$ [3+6] (76 Ash)
 $\rightarrow X(z) = \frac{z^2 [1 - \frac{3}{2}z^{-1}](1 + z^{-1})(1 - z^{-1})}{z^2 - 1.5z - 1}, |z| < 0.5$ [3+3] (74 Ash)
 $\rightarrow X(z) = \frac{2z^2 + 2z^2 + 3z + 5}{z^2 - 0.1z - 0.2}, |z| < 0.4$ [1+3+5] (**74 Ch,73 Ch**)
 $\rightarrow X(z) = \frac{z}{(z - 0.4)(z + 1.5)^2}, |z| < 0.4$ [1+5] (73 Shr)
 $\rightarrow X(z) = \frac{1}{(z - 0.5)(z + 2)}, 0.5 < |z| < 2, |z| < 0.5, |z| > 2$ [1+5] (**71 Ch**)
 $\rightarrow X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + 1.5z^{-1} + 0.5z^{-2}}, |z| > 1$ [6] (71 Shr)
 $\rightarrow X(z) = \frac{z}{(z - 1)(z - 2)^2}, |z| < 1$ [1+5] (70 Ash)

3 Analysis of LTI Systems in Frequency Domain

(6 Hours)

3.1 Pole-zero plot and magnitude/phase response

- Plot the pole-zero in the z-plane and draw the magnitude response (not to scale) of the system described by the given difference equation:
 - $\rightarrow y(n) = x(n) + 0.8x(n-1) + 0.8x(n-2) + 0.49y(n-2)$ [3+7] (80 Bh)
 - $\rightarrow y(n) = 0.67x(n) - 0.3x(n-1) + 2.75y(n-1)$ [3+7] (81 Ba); phase response [2+8] (76 Ash)
 - $\rightarrow y[n] - 0.35y[n-1] + 0.25y[n-2] = x[n] - 0.75x[n-1]$ [3+7] (79 Bh)
 - $\rightarrow y[n] - 0.3y[n-1] + 0.2y[n-2] = x[n] - 0.5x[n-1]$ [3+7] (79 Ba)
 - $\rightarrow y[n] - 0.4y[n-1] + 0.25y[n-2] = x[n] - 0.4x[n-1]$ [2+4] (78 Bh)
 - $\rightarrow y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.5x[n-1] + 0.6x[n-2] + 0.8x[n-3]$ [3+7] (74 Ash)
 - $\rightarrow y[n] - 0.5y[n-1] + 0.3y[n-2] = x[n] + 0.7x[n-1]$ [6] (72 Ka)
 - $\rightarrow y[n] - 0.4y[n-1] + 0.1y[n-2] = x[n] + 0.6x[n-1]$ [2+4] (72 Ch)
 - $\rightarrow y[n] - 0.3y[n-1] + 0.225y[n-2] = x[n] + 0.5x[n-1]$ [6] (70 Ch)
 - $\rightarrow y[n] - 0.4y[n-1] + 0.2y[n-2] = x[n] + 0.1x[n-1] - 0.06x[n-2]$ [2+2+6] (69 Ch)
 - $\rightarrow y(n) - 0.3y(n-1) = 2x(n-2) + 0.7x(n-1) + 4x(n)$ [2+7] (75 Ch)
- Map the poles and zeros in the z-plane and plot the magnitude response for a system with given pole/zero locations:
 - $\rightarrow$ poles $0.45 \pm j1.6$, zeros $0.58 \pm j2.06$ [4+6] (80 Ba)
 - $\rightarrow$ poles $0.45 \pm j1.06$, zeros $0.58 \pm j2.06$ [2+6/2+8] (76 Ch, 75 Ash)
 - $\rightarrow$ poles $0.45 - 0.77i$ and $-2 \pm 0.3i$ [2+8] (74 Ch)
 - $\rightarrow$ poles $0.45 + 0.77i$, $2 \pm 0.7i$, zeros $1.2 \pm 0.43i$ [2+8] (73 Ch)
 - $\rightarrow$ poles $0.45 + 0.77i$, $-2 \pm 0.3i$, zeros $1.2 \pm 3i$ [2+8] (71 Ch)
 - $\rightarrow y(n) = x(n) + x(n-2)$: poles/zeros + magnitude + causal & stable? [2+6+2] (70 Ash)

3.2 FIR vs IIR, LCCDE, linear phase, stability/causality

- Differentiate between FIR system and IIR system. [4+6] (80 Ba)
- State linear constant coefficient difference equation and the corresponding system function. [3+7] (73 Shr) [2+3+5] (71 Shr)
- Determine the zero-input response for the second order system $y[n] - 3y[n-1] - 4y[n-2] = x[n]$. [4] (78 Bh)
- Describe stability and causality characteristics of an LTI system in terms of Impulse Response and ROC of its transfer function with examples. [4+3] (76 Ash) [4] (72 Ch)
- What is the condition satisfied by a linear phase FIR filter? Show that the filter with $h(n) = \{-1, 0, 1\}$ is a linear phase filter. [2+4] (70 Ash)

4 Discrete Filter Structures

(8 Hours)

4.1 Direct form and cascade realizations

1. Obtain the Direct Form I and Direct Form II realization of the given system:

$$\begin{aligned} \rightarrow y[n] &= -0.25y[n-2] + x[n] + 0.4x[n-1] + 0.5x[n-2] & [2+2] \text{ (79 Bh)} \\ \rightarrow y[n] - 0.75y[n-1] - 0.25y[n-2] &= x[n] + 0.5x[n-1] & [4] \text{ (79 Ba)} \\ \rightarrow 3y[n] + y[n-1] + 2y[n-4] &= 2x[n] + x[n-3] & [5] \text{ (74 Ch)} \\ \rightarrow y(n) &= -0.1y(n-1) + 0.2y(n-2) + 3x(n) + 3.6x(n-2) + 0.6x(n-2) & [5] \text{ (72 Ch)} \\ \rightarrow \text{Direct Form II only: } y(n) &= -0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.252x(n-2) & [4] \text{ (72 Ka, 69 Ch)} \end{aligned}$$

2. Realize the given system in cascade form of second-order sections (signal flow graph):

$$\begin{aligned} \rightarrow H(z) &= 10(1-0.25z^{-1})(1-0.667z^{-1})(1+2z^{-1})/[(1-0.75z^{-1})(1-0.125z^{-1})(1-(0.5+j0.5)z^{-1})(1-(0.5-j0.5)z^{-1})] & [5] \text{ (81 Ba)} \\ \rightarrow H(z) &= \{(1-0.4z^{-1})(1+0.2z^{-1})(1-0.3e^{j\pi/6}z^{-1})(1-0.3e^{-j\pi/6}z^{-1})\}/\{(1-0.5e^{j\pi/3}z^{-1})(1-0.5e^{-j\pi/3}z^{-1})(1+0.7e^{j\pi/4}z^{-1})(1+0.7e^{-j\pi/4}z^{-1})\} & [4] \text{ (80 Ba)} \\ \rightarrow \text{Similar 2nd-order cascade with } e^{j2n\pi/5} \text{ etc.} & & [4] \text{ (76 Ch)} \\ \rightarrow y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] - x[n] - 2x[n-1] &= 0 & [4] \text{ (70 Ch)} \end{aligned}$$

4.2 Lattice / lattice-ladder structures

1. Draw the lattice (or lattice-ladder) structure for the given system function (and check stability where asked):

$$\begin{aligned} \rightarrow H(z) &= 1/(1-0.2z^{-1}+0.4z^{-2}+0.6z^{-3}) & [10] \text{ (80 Bh)} \\ \rightarrow \text{FIR } H(z) &= 1 + (13/24)z^{-1} + (5/8)z^{-2} + (1/3)z^{-3}, \text{ check stability} & [5] \text{ (81 Ba); Direct + Lattice} \\ & [3+7] \text{ (78 Bh)} \\ \rightarrow \text{lattice-ladder } H(z) &= (2-0.7z^{-1}+0.5z^{-2})/(1-0.3z^{-1}+0.25z^{-2}) & [6] \text{ (80 Ba)} \\ \rightarrow \text{lattice-ladder } H(z) &= (1-0.4z^{-1}+0.25z^{-2})/(1-0.3z^{-1}+0.5z^{-2}), \text{ stability} & [6+1] \text{ (79 Ba)} \\ \rightarrow \text{lattice-ladder } H(z) &= (0.5-2z^{-1}+3z^{-2})/(1-0.5z^{-1}-0.7z^{-2}+0.3z^{-3}) & [6+4] \text{ (76 Ch)} \\ \rightarrow \text{lattice-ladder } H(z) &= (0.7-1.5z^{-1}+0.5z^{-2})/(1-0.5z^{-1}-0.7z^{-2}+0.3z^{-3}) & [6+3] \text{ (76 Ash)} \\ \rightarrow H &= 1/(3 + (39/24)z^{-1} + (15/8)z^{-2} + (3/9)z^{-3}); \text{ also represent } 5/8, -5/8 \text{ in sign-magnitude, 1's} \\ & \text{and 2's complement} & [7+3] \text{ (75 Ash)} \\ \rightarrow H(t) &= 1/(1 + \frac{2}{3}z^{-1} + \frac{5}{8}z^{-2} + \frac{2}{3}z^{-3} + z^{-4}) & [9] \text{ (75 Ch)} \\ \rightarrow \text{FIR } H(z) &= 1 + 2z^{-1} + z^{-2} & [5] \text{ (72 Ch)} \\ \rightarrow \text{FIR } H(z) &= A_3(z) = 1 + (52/96)z^{-1} + (25/40)z^{-2} + (1/3)z^{-3} & [5] \text{ (74 Ch)} \\ \rightarrow \text{Direct + Lattice, FIR } H(z) &= 1 + 0.7z^{-1} + 1.2z^{-2} - z^{-3} & [3+7] \text{ (74 Ash)} \\ \rightarrow \text{Direct + Lattice, } H(z) &= 2 + 1.8z^{-1} - 1.6z^{-2} + z^{-3} & [3+7] \text{ (73 Shr)} \\ \rightarrow \text{Direct + Lattice, } H(z) &= 1 + \frac{1}{3}z^{-1} + \frac{9}{8}z^{-2} + \frac{4}{3}z^{-3} + z^{-4} \text{ over denominator} & [10] \text{ (73 Ch)} \\ \rightarrow \text{FIR } H(z) &= 1 + 2z^{-1} - 3z^{-2} + 4z^{-3} & [6] \text{ (72 Ka, 69 Ch)} \\ \rightarrow \text{FIR } H(z) &= 1 + 3.1z^{-1} + 5.5z^{-2} + 4.2z^{-3} + 2.3z^{-4} & [6] \text{ (70 Ch)} \\ \rightarrow \text{IIR } H(z) &= 1/(1-0.01z^{-1}-0.23z^{-2}+0.5z^{-3}), \text{ stability} & [4+2+1] \text{ (71 Ch)} \\ \rightarrow \text{pole-zero } H(z) &= (0.5+2z^{-1}+0.6z^{-2})/(1-0.3z^{-1}+0.4z^{-2}) & [6] \text{ (70 Ash)} \end{aligned}$$

2. Given a 3-stage lattice filter for an all-zero/all-pole polynomial with coefficients $K_1 = 1/4, K_2 = 1/2, K_3 = 1/3$: obtain the system function and FIR filter coefficients. [6] (79 Bh) [5] (71 Shr)

4.3 Quantization effects, limit cycles, dead band

1. What is the limit cycle effect in a recursive system? Describe with an example showing how it occurs.
[3] (**71 Ch**) [1+3] (70 Ash)
2. What is the importance of quantization in DSP? Which is better, rounding or truncation? Explain limit cycles in a recursive system and define dead band.
[1+1+2+1] (71 Shr)

5 FIR Filter Design

(6 Hours)

5.1 Window method and Gibbs phenomenon

1. Describe how a digital FIR filter is designed by the window method (discuss characteristics of different window functions). [4+4] (**70 Ch**) [5+3] (**72 Ch**)
2. Design a symmetric/linear-phase FIR LPF using a suitable window:
 - |→ $H_d(\omega) = e^{-j\omega\tau}$, $|\omega| \leq \omega_c$; length 7, $\omega_c = 1$ rad; Hanning window [10] (**80 Bh**)
 - |→ $\omega_p = 0.2\pi$, $\omega_s = 0.45\pi$, $\alpha_s = 51$ dB, any window [5+3] (79 Ba)
 - |→ $\omega_p = 0.3\pi$, $\omega_s = 0.5\pi$, $\alpha_s = 40$ dB, suitable window [8] (**71 Ch, 69 Ch**)
 - |→ first six coefficients; $\omega_p = 0.2\pi$, $\omega_s = 0.5\pi$, $\alpha_s = 41$ dB (Why is Kaiser better?) [2+6] (74 Ash)
 - |→ Hanning window; $\omega_p = 0.25\pi$, $\omega_s = 0.35\pi$, main-lobe width $8\pi/M$ [9] (70 Ash)
3. Explain Gibbs phenomenon / oscillation and how it arises with the rectangular window (and how it can be minimized). [3+3/5/6] (81 Ba, 80 Ba, 73 Shr, 71 Shr)

5.2 Kaiser window design

1. Design a linear phase FIR filter using the Kaiser window for the given specifications (determine ripple, β , and length $M + 1$ where asked):
 - |→ $0.99 \leq |H| \leq 1.01$, $0 \leq |w| \leq 0.3\pi$; $|H| \leq 0.01$, $0.35\pi \leq |w| \leq \pi$ [10] (81 Ba)
 - |→ $|H| \leq 0.01$, $0.21\pi \leq |w| \leq \pi$; $0.95 \leq |H| \leq 1.05$, $0 \leq |w| \leq 0.19\pi$ (ripple, atten, length) [2+2+2] (**80 Bh**)
 - |→ $0.99 \leq |H| \leq 1.01$, $0 \leq w \leq 0.016\pi$; $|H| \leq 0.01$, $0.08\pi \leq w \leq 2\pi$ [2+8] (80 Ba)
 - |→ $0.899 \leq |H| \leq 1$, $|w| \leq 0.2\pi$; $|H| \leq 0.01$, $0.4\pi \leq w \leq \pi$ [2+8] (**79 Bh**)
 - |→ $|H| \leq 0.01$, $0 \leq w \leq 0.25\pi$; $0.95 \leq |H| \leq 1.05$, $0.35\pi \leq w \leq 0.6\pi$; $|H| \leq 0.01$, $0.65\pi \leq w \leq \pi$ [8] (**78 Bh**); with min length $M + 1$ and β [9+3] (**74 Ch**)
 - |→ $0.99 \leq |H| \leq 1.01$, $0 \leq w \leq 0.19\pi$; $|H| \leq 0.01$, $0.21\pi \leq w \leq \pi$ (+ optimum-filter flowchart) [8+4] (75 Ash) [6+8] (72 Ka)
 - |→ $0.99 \leq |H| \leq 1.01$, $0 \leq w \leq 0.16\pi$; $|H| \leq 0.01$, $0.18\pi \leq w \leq 2\pi$ (+ Remez flowchart) [2+4+4] (76 Ash)
 - |→ $0.98 \leq |H| \leq 1.02$, $0 \leq w \leq 0.9\pi$; $|H| \leq 0.01$, $0.14\pi \leq w \leq \pi$ [8] (**73 Ch**)

5.3 Optimum filter / Remez exchange algorithm

1. What is an optimum filter? Show the mathematical expression / draw the flowchart of the Remez exchange algorithm for FIR filter design (with derivation where asked). [1+5/1+6/2+6/7/9] (**75 Ch, 72 Ch, 70 Ch, 69 Ch**, 76 Ash, 73 Shr, 71 Shr, **73 Ch**)
 - |→ Describe the Remez exchange algorithm with flowchart. [5/1+6] (**79 Bh**, 79 Ba)

6 IIR Filter Design

(6 Hours)

6.1 Bilinear transformation (Butterworth)

- Design a digital IIR/Butterworth LPF using the bilinear transformation method for the given specifications:
 - $\rightarrow 0.9 \leq |H| \leq 1, 0 \leq w \leq \pi/2; |H| \leq 0.2, 3\pi/4 \leq w \leq \pi; f_s = 1$ [12] (80 Bh)
 - $\rightarrow 0.6 \leq |H| \leq 1, 0 \leq \omega \leq 0.35\pi; |H| \leq 0.1, 0.7\pi \leq \omega \leq \pi; T = 0.1$ s (bilinear Vs impulse invariance) [2+10] (81 Ba)
 - $\rightarrow$ passband edge 0.26π , deviation 0.99 dB; max gain -14.99 dB at $0.58\pi; f_s = 0.5$ [11] (80 Ba)
 - $\rightarrow W_p = 0.25\pi, W_s = 0.55\pi, \delta_p = 0.11, \delta_s = 0.21, f_s = 0.5$; then convert to HPF $W'_p = 0.45\pi$ [11+4] (79 Bh, 71 Ch, 69 Ch)
 - $\rightarrow$ passband edge 0.24π , deviation 0.98 dB; max gain -14.95 dB at $0.57\pi; f_s = 0.5$ (compare with impulse invariance) [11+3] (79 Ba)
 - $\rightarrow 0.99 \leq |H| \leq 1.01 \dots$ (specs) (various)
 - $\rightarrow 0.8 \leq |H| \leq 1, 0 \leq W \leq 0.2\pi; |H| \leq 0.2, 0.6\pi \leq W \leq \pi$ [10] (75 Ch, 70 Ch)
 - $\rightarrow 0.707 \leq |H| \leq 1, 0 \leq w \leq \pi/2; |H| \leq 0.2, 3\pi/4 \leq w \leq \pi; T = 1$ s (realize convenient form) [11+4] (73 Ch)
 - $\rightarrow 0.89125 \leq |H| \leq 1, 0 \leq \omega \leq 0.2\pi; |H| \leq 0.17783, 0.3\pi \leq \omega \leq \pi$ [12] (73 Shr)
 - $\rightarrow$ passband peak-to-peak ripple ≤ 1 dB, passband edge 1.2 kHz; stopband atten ≥ 40 dB, edge 2.5 kHz; $f_s = 8$ kHz [10] (76 Ch)
 - $\rightarrow$ passband const 0.7 dB below 0.15π ; stopband ≥ 14 dB, 0.6π to π [12] (75 Ash, 74 Ch)
 - $\rightarrow \omega_p = 0.22\pi, \omega_s = 0.54\pi, \delta_p = 0.11, \delta_s = 0.22, f_s = 0.5$ [12] (76 Ash)
 - $\rightarrow \omega_p = 0.27\pi, \omega_s = 0.58\pi, \delta_p = 0.11, \delta_s = 0.21, f_s = 0.5$ [15] (74 Ash)
 - $\rightarrow$ passband edge 0.25π , deviation 1 dB; max gain -15 dB at $0.45\pi; f_s = 1$ [15] (72 Ka)
 - $\rightarrow$ passband edge 0.2π , deviation 1 dB; max gain -15 dB at $0.4\pi; f_s = 1$ [15] (71 Shr)

6.2 Impulse invariance method

- Design a Butterworth LPF using the Impulse Invariance Method:
 - $\rightarrow$ passband/stopband 200 Hz / 500 Hz, atten 5 dB / 12 dB, $f_s = 5000$ Hz; explain pre-warping and why it is necessary [12+3] (78 Bh)
 - $\rightarrow (+)$ of bilinear over impulse invariance; 200/500 Hz, $-5/-12$ dB, $f_s = 5$ kHz (IIM) [3+12] (72 Ch)
 - $\rightarrow$ passband edge 0.2π , deviation 0.5 dB; max gain -15 dB at $0.35\pi; f_s = 1$ [13] (70 Ash)
- Explain briefly the bilinear transformation method of IIR filter design / its advantages over impulse invariance. [3] (73 Shr)

6.3 Spectral (digital domain) transformation

- Describe digital-domain spectral transformation features and parameters for low-pass to high-pass conversion in IIR filter design. [4] (80 Ba)
- Why is spectral transformation required? [2] (70 Ash)
- A digital LPF with $w_c = 0.2575\pi$ is $H(z) = (0.1 + 0.4z^{-1}) / (1 - 0.6z^{-1} + 0.1z^{-2})$; design a digital HPF with $w'_c = 0.3567\pi$. [5] (70 Ch)

7 Discrete Fourier Transform (DFT and FFT)

(7 Hours)

7.1 DFT — definition, properties, DFT vs DTFT

1. Why do we need DFT / FFT (although we have DTFT)? Differentiate between DFT and DTFT; write the computational complexity of DFT and FFT. [2-2+5] (**79 Bh, 78 Bh, 71 Ch**, 79 Ba, 80 Ba)
2. State and prove the circular convolution property of DFT. [2+2+4] (**70 Ch**)
3. Define zero padding / padding zones. [1] (**74 Ch, 74 Ash, 70 Ch**)

7.2 FFT computation (DIT / DIF)

1. Find the N -point DFT of the given sequence using the DIT-FFT or DIF-FFT algorithm (draw the butterfly diagram where asked):
 - 8-pt DIF, $x(n) = \{2, 1, 2, 1, 1, 2, 1, 2\}$ [8] (**80 Bh**)
 - 4-pt DIT, $x(n) = \{2, 2, 4\}$ (Why DIT-FFT better than direct DFT?) [2+6] (81 Ba)
 - 8-pt DIF, $x[n] = \{1, 1, 0, 0, 1, 1, 2\}$ [2+6] (80 Ba, **70 Ch**)
 - 8-pt radix-2 DIT, $x[n] = u[n] - u[n-4]$ / general [2+6/7] (**79 Bh, 78 Bh**)
 - 8-pt DIF, $x[n] = \{1, 2, 4, 3, 5, -1, 3\}$ [2+6] (79 Ba)
 - 8-pt DIF, $x[n] = \{1, 2, 3, 3, 5, 0, 4, 6\}$ [7] (**76 Ch**)
 - 8-pt DIF, $x[n] = \{1, 2, 3, 3, 5, 1, 4, 2\}$ [2+8] (76 Ash)
 - $X(3), X(5)$ of $x[n] = \{1, -2, 3, 2\}$ (DIT-FFT) [2+8] (**75 Ch**)
 - butterfly 8-pt DIT, $x[n] = n + 1$ [3+7] (75 Ash)
 - butterfly radix-2 DIF, $X(2), X(1)$ of $x[n] = \{1.5, -1, 1.8, 0.6, 3, 1.7\}$ [3+7] (**74 Ch**)
 - 8-pt DIF, $x[n] = \{1/2, 1/2, 1/2, 1/2, 0, 0, 0, 0\}$ [7] (74 Ash)
 - 8-pt DIF, $x[n] = \{1, -1, 3, 2, 1, 1, 3, -2\}$ [2+6] (**73 Ch**)
 - 8-pt, $x[n] = \{1, -1, 2, 2, 1, 1, 2, 2\}$ [2+6] (73 Shr)
 - DIT, $x[n] = \{1, 1, 2, 4, 3, 1, 2, 1\}$ [8] (**72 Ch**)
 - 8-pt DIT, $x[n] = \{1, 1, 0, 1, 0, 1, 2\}$ [7] (72 Ka)
 - butterfly $X(7)$, 8-pt DIT, $x(n) = \{1, 0, 0, 0, 0, 0, 0, 0\}$ [2+6] (**71 Ch**, 70 Ash)
 - DIT butterfly + spectrum, $x = \{1, 1, 2, 0, 1, 2, 0, 1\}$ [6+2] (71 Shr)
 - 8-pt DIF, $x[n] = \{1, 1, 2, 2, 1, 1, 2, 1\}$ [2+7] (**69 Ch**)

7.3 Circular convolution (direct and via DFT: $x_3 = \text{IDFT}(X_1 X_2)$)

1. Compute the circular convolution of the given sequences (directly or using DFT):
 - $x_1 = \{1, 2, 3, 1\}$, $x_2 = \{4, 3, 2, 2\}$ [6/2+5] (**80 Bh**, 80 Ba)
 - $x = \{1, 2, 4, 5\}$, $h = \{2, 1, 6, 8\}$ [6] (81 Ba)
 - $x_1 = \{1, -1, -2, 3, -1\}$, $x_2 = \{1, 2, 3\}$ [7] (79 Ba)
 - $x_1 = \{2, 1, 2, 1\}$, $x_2 = \{1, 2, 3, 4\}$ [2+6] (**78 Bh**)
 - $x_1 = \{1, 2, -1, 1\}$, $x_2 = \{1, 3, 5, 7\}$ [1+5] (75 Ash)
 - linear Vs circular; $X_1 = \{0, 0, 1, 1\}$, $X_2 = \{1, 1, 1, 1\}$ [3+7] (**75 Ch**)
 - linear conv via circular with zero padding; $x = \{1, 1, 1, 1\}$, $h = \{2, 3\}$ [1+5] (**74 Ch**)
 - $h(n) = \{1, 2, 1, -1, 1\}$, $x = \{1, 2, 3, 1\}$ [7] (**72 Ch**)
 - circular conv; $x = \{1, 2\}$, $y = u[n] - u[n-4]$ [2+5] (**71 Ch**, 70 Ash)
2. Find $x_3[n]$ if $X_3(k) = X_1(k)X_2(k)$ (product of DFTs $\Rightarrow$ circular convolution):
 - 4-pt: $x_1 = \{1, 2, -2\}$, $x_2 = \{1, 2, 3, -1\}$ [5/6] (**76 Ch, 69 Ch**)
 - 5-pt: $x_1 = \{1, -2, 5, 1, 2\}$, $x_2 = \{1, 2, -3, -2\}$ [7] (**73 Ch**)
 - 5-pt: $x_1 = \{1, -2, 2, 1, 4\}$, $x_2 = \{2, 1, -3, -1\}$ [7] (73 Shr)

$\rightarrow x_1 = \{1, 2, 4\}, x_2 = \{-1, 2, 3, 1\}$	[2+6] (72 Ka)
$\rightarrow x_1 = \{1, 0, 0, 1\}, x_2 = \{2, 0, 2\}$	[7] (79 Bh)
$\rightarrow$ 5-pt: $x_1 = \{1, 2, 0, 1, -2\}, x_2 = \{1, 0, 1, 1, 2\}$ (zero padding)	[1+7] (74 Ch)
$\rightarrow$ System $x[n] = \{1, 2, 3, 4\}, h[n] = \{1, 3, 5, 7\}$; output $y[n]$ if $G[k] = X[k]H[k]$	[7] (71 Shr)